

Polynomial Long Division

Answer Key

Algebra 2

Quick Answers

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|---|---|
| 1. $x + 4$ | $x = 2 = 7$ |
| 2. $x + 3 + \frac{3}{x+2}$ | 8. $2x^2 - 5x + 7 - \frac{16}{x+3}$ |
| 3. $2x^2 + 3x - 2$ | 9. $x + 4 - \frac{4x+2}{x^2+1}$ |
| 4. $x^2 + 2x + 4$ | 10. $2x^3 + x^2 + 2x + 5 + \frac{4}{x-2}$ |
| 5. $3x - 2 + \frac{2}{x-1}$ | 11. the quotient = $x^2 + x - 6$, the complete factorization = $(x - 1)(x + 3)(x - 2)$ |
| 6. $x^2 - 4$ | 12. the value of k = -3 , — |
| 7. the quotient and remainder = $x^2 + 4x + 3 + \frac{7}{x-2}$, the value of the dividend at | |
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What is verified

14 of 15 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-APR.B.3 – problems 7, 11, 12

HSA-APR.D.6 – problems 1, 2, 3, 4, 5, 6, 8, 9, 10

Difficulty 1–5 of 5 across 15 tagged checks.

1. Divide $\frac{x^2 + 7x + 12}{x + 3}$.

Solution: $x^2 \div x = x$; $x(x + 3) = x^2 + 3x$; subtracting leaves $4x + 12$. Then $4x \div x = 4$; $4(x + 3) = 4x + 12$; subtracting leaves 0.

$x + 4$

2. Divide $\frac{x^2 + 5x + 9}{x + 2}$.

Solution: First step gives x and leaves $3x + 9$; second step gives 3 and leaves $9 - 6 = 3$. Degree 0 is below the divisor's degree 1, so the division stops there and 3 is the remainder.

$x + 3 + \frac{3}{x + 2}$

3. Divide $\frac{2x^3 - 3x^2 - 11x + 6}{x - 3}$.

Solution: The three steps give $2x^2$, then $3x$, then -2 :

$$\begin{array}{ll} 2x^2(x - 3) = 2x^3 - 6x^2, & \text{leaving } 3x^2 - 11x + 6 \\ 3x(x - 3) = 3x^2 - 9x, & \text{leaving } -2x + 6 \\ -2(x - 3) = -2x + 6, & \text{leaving } 0 \end{array}$$

$2x^2 + 3x - 2$

4. Divide $\frac{x^3 - 8}{x - 2}$.

Solution: Write the dividend as $x^3 + 0x^2 + 0x - 8$ first — the placeholders are what keep the columns aligned. The steps give x^2 , then $2x$, then 4 , with nothing left over. This is the difference-of-cubes factorisation seen from the other side.

$x^2 + 2x + 4$

5. Divide $\frac{3x^2 - 5x + 4}{x - 1}$.

Solution: $3x^2 \div x = 3x$; $3x(x - 1) = 3x^2 - 3x$, leaving $-2x + 4$. Then $-2(x - 1) = -2x + 2$, leaving 2 .

$3x - 2 + \frac{2}{x - 1}$

6. Divide $\frac{x^4 - 5x^2 + 4}{x^2 - 1}$.

Solution: Write the dividend as $x^4 + 0x^3 - 5x^2 + 0x + 4$. Dividing leading terms, $x^4 \div x^2 = x^2$, and $x^2(x^2 - 1) = x^4 - x^2$, leaving $-4x^2 + 4$. Then $-4(x^2 - 1) = -4x^2 + 4$, leaving 0 .

$x^2 - 4$

7. $\frac{x^3 + 2x^2 - 5x + 1}{x - 2}$. (a) Quotient and remainder. (b) Value of the dividend at $x = 2$.

Solution (a): The steps give x^2 , then $4x$, then 3 , and the last subtraction leaves 7 .

$x^2 + 4x + 3 + \frac{7}{x - 2}$

Solution (b): $8 + 8 - 10 + 1$.

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The two answers agree, and they always will: that is the remainder theorem. Dividing by $x - a$ leaves the remainder $P(a)$.

8. Divide $\frac{2x^3 + x^2 - 8x + 5}{x + 3}$.

Solution: Dividing by $x + 3$ means $a = -3$. The steps give $2x^2$ (leaving $-5x^2 - 8x + 5$), then $-5x$ (leaving $7x + 5$), then 7 , and the last subtraction leaves $5 - 21 = -16$.

$$2x^2 - 5x + 7 - \frac{16}{x+3}$$

9. Divide $\frac{x^3 + 4x^2 - 3x + 2}{x^2 + 1}$.

Solution: The divisor has degree 2, so the division stops as soon as what remains has degree 1 or less.

$$\begin{array}{ll} x(x^2 + 1) = x^3 + x, & \text{leaving } 4x^2 - 4x + 2 \\ 4(x^2 + 1) = 4x^2 + 4, & \text{leaving } -4x - 2 \end{array}$$

Degree 1 is below degree 2, so stop.

$$x + 4 - \frac{4x + 2}{x^2 + 1}$$

10. Divide $\frac{2x^4 - 3x^3 + x - 6}{x - 2}$.

Solution: The x^2 term is missing, so write $2x^4 - 3x^3 + 0x^2 + x - 6$. The four steps give $2x^3$, x^2 , $2x$, 5, and the last subtraction leaves $-6 + 10 = 4$.

$$2x^3 + x^2 + 2x + 5 + \frac{4}{x - 2}$$

11. $\frac{x^3 - 7x + 6}{x - 1}$. (a) Quotient. (b) Factor $x^3 - 7x + 6$ completely.

Solution (a): With the placeholder, the dividend is $x^3 + 0x^2 - 7x + 6$. The steps give x^2 , then x , then -6 , and nothing is left.

Solution (b): A zero remainder means $x - 1$ is a factor, so $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6)$, and the quadratic factors further into $(x + 3)(x - 2)$.

$$x^2 + x - 6$$

$$(x - 1)(x + 3)(x - 2)$$

12. $\frac{x^3 + kx^2 - 4x + 12}{x - 2}$ leaves no remainder. (a) Find k . (b) Explain the shortcut.

Solution (a): By the remainder theorem the remainder is the dividend evaluated at $x = 2$:

$$8 + 4k - 8 + 12 = 4k + 12.$$

Setting that to 0 gives $4k = -12$.

$$k = -3$$

Part (b) — instructor-judged. A full-credit explanation states the remainder theorem — dividing by $x - 2$ leaves a remainder equal to the value of the dividend at 2 — and then says that “no remainder” therefore means “the value at 2 is zero”, which is a single linear equation in k instead of a full division. A student who writes the substitution and the resulting equation has shown it. Describing the long division the theorem was meant to replace earns nothing.